

| Assessment | Test | Domain        | Skill               | Difficulty                                          |
|------------|------|---------------|---------------------|-----------------------------------------------------|
| SAT        | Math | Advanced Math | Nonlinear functions | Medium <div><div></div><div></div><div></div></div> |

Question

For the polynomial function  $f$ , the graph of  $y = f(x)$  in the  $xy$ -plane passes through the points  $(-5, 0)$ ,  $(1, 0)$ , and  $(4, 0)$ . Which of the following must be a factor of  $f(x)$ ?

Answer

- A.  $x + 1$
- B.  $x + 4$
- C.  $x - 1$
- D.  $x - 5$

Correct Answer: C

Rationale

Choice C is correct. It's given that  $f$  is a polynomial function and the graph of  $y = f(x)$  in the  $xy$ -plane passes through the points  $(-5, 0)$ ,  $(1, 0)$ , and  $(4, 0)$ . A point in the form  $(a, 0)$  on the graph of  $f(x)$  means that  $x - a$  is a factor of  $f(x)$ . Since the graph of the function  $f$  passes through the point  $(1, 0)$ ,  $x - 1$  must be a factor of  $f(x)$ .

Choice A is incorrect. This is a factor if the graph of  $y = f(x)$  passed through  $(-1, 0)$ .

Choice B is incorrect. This is a factor if the graph of  $y = f(x)$  passed through  $(-4, 0)$ .

Choice D is incorrect. This is a factor if the graph of  $y = f(x)$  passed through  $(5, 0)$ .